A Branch-Safe Spectral and Scattering Atlas for a Side-Coupled Fano–Anderson Impurity

Route-A source-local certificate HCS-C345

3 September 2026

Abstract

We solve the infinite nearest-neighbour chain with one discrete state side-coupled at the origin. A physical-sheet Schur function gives exactly two simple bound states for every nonzero coupling, one on each side of the free band. Squaring the secular equation can create extra real roots; an explicit sign filter prevents their promotion to eigenvalues. Reflection parity and a cyclic eventually-free Jacobi representation yield the complete spectral type, band multiplicity, impurity density, residue weights, and mass normalization. Stationary matching gives exact unitary reflection and transmission laws and the if-and-only-if location of the Fano zero.

1 Frozen Hamiltonian and physical sheet

Let

$$\mathcal{H} = \ell^2(\mathbb{Z}) \oplus \mathbb{C}|d\rangle, \qquad J > 0, \quad \varepsilon \in \mathbb{R}, \quad g \in \mathbb{R},$$

and define

$$(Hu)_n = J(u_{n+1} + u_{n-1}) + g u_d \delta_{n0}, \tag{1}$$

$$(Hu)_d = \varepsilon u_d + g u_0. \tag{2}$$

This is a bounded self-adjoint operator on all of $\mathcal{H}$. Write H_0 for the free chain. Fourier transform makes H_0 multiplication by $2J\cos k$, so

$$\sigma(H_0) = \sigma_{\mathrm{ac}}(H_0) = [-2J, 2J]$$

with multiplicity two in the open band.

Fix the analytic branch

$$S(z) = \sqrt{z^2 - 4J^2}, \qquad S(z)/z \longrightarrow 1 \quad (z \longrightarrow \infty).$$

Thus $S(E) > 0$ for $E > 2J$ and $S(E) < 0$ for $E < -2J$. A contour integral, with the normalization checked at infinity, gives

$$m(z) = \langle 0, (z - H_0)^{-1} 0 \rangle = \frac{1}{S(z)}. \tag{3}$$

Block inversion at $|d\rangle$ now yields

$$G_{dd}(z) = \langle d, (z - H)^{-1} d \rangle = \frac{1}{D(z)}, \qquad D(z) = z - \varepsilon - \frac{g^2}{S(z)}. \tag{4}$$

Theorem 1 (Two physical-sheet bound states)**.** *If $g \neq 0$, H has exactly two exterior eigenvalues. They are simple and are the unique solutions*

$$E_+ - \varepsilon - \frac{g^2}{\sqrt{E_+^2 - 4J^2}} = 0, \qquad E_+ > \max(2J, \varepsilon), \tag{5}$$

$$E_- - \varepsilon + \frac{g^2}{\sqrt{E_-^2 - 4J^2}} = 0, \qquad E_- < \min(-2J, \varepsilon). \tag{6}$$

Squaring produces

$$(E - \varepsilon)^2 (E^2 - 4J^2) - g^4 = 0, \tag{7}$$

but a real root of (7) is an eigenvalue only when it also satisfies the corresponding inequality in (5) or (6).

Proof. For $E > 2J$ put

$$D_+(E) = E - \varepsilon - \frac{g^2}{\sqrt{E^2 - 4J^2}}.$$

It tends to $-\infty$ at $2J+$ and to $+\infty$ at infinity, while

$$D_+'(E) = 1 + \frac{g^2 E}{(E^2 - 4J^2)^{3/2}} > 0.$$

Hence it has exactly one simple zero. The equation itself forces $E_+ > \varepsilon$. Below the band use

$$D_-(E) = E - \varepsilon + \frac{g^2}{\sqrt{E^2 - 4J^2}}.$$

It tends to $-\infty$ at minus infinity and to $+\infty$ at $-2J-$, and

$$D'_-(E) = 1 - \frac{g^2 E}{(E^2 - 4J^2)^{3/2}} > 0.$$

The unique zero is simple and satisfies $E_- < \varepsilon$.

At an exterior zero, equations (1)–(2) give the square-summable vector $u_{\mathbb{Z}} = g u_d (E - H_0)^{-1}|0\rangle$. Conversely, an exterior eigenvector cannot have $u_d = 0$, because H_0 has no exterior eigenvalue; its impurity equation therefore gives $D(E) = 0$. This establishes the pole/eigenvalue equivalence. Squaring loses the sign of $S(E)$ and the sign of $E - \varepsilon$, which proves the stated filter. ☐

The phrase *branch-safe Schur owner* marks the baseline manuscript: the quartic is an algebraic candidate ledger, never an unfiltered spectrum.

2 Full spectral measure

Let reflection send u_n to u_{-n}. The odd and even subspaces reduce H. On the odd subspace, $u_0 = u_d = 0$ and H is the free Dirichlet half-line Jacobi matrix. It has simple purely absolutely continuous spectrum on $[-2J, 2J]$.

On the even-plus-impurity subspace, use the ordered basis

$$|d\rangle,\ |0\rangle, \qquad |e_n\rangle = \frac{|n\rangle + |-n\rangle}{\sqrt{2}} \quad (n \geq 1).$$

The Jacobi diagonal is $\varepsilon, 0, 0, \ldots$ and its successive off-diagonals are $g, \sqrt{2}J, J, J, \ldots$. If $g \neq 0$, $|d\rangle$ is cyclic. Its scalar resolvent is (4), so that scalar measure therefore determines the entire even spectral type.

Let μ_d be that scalar spectral measure. Our convention is

$$G_{dd}(z) = \int_{\mathbb{R}} \frac{d\mu_d(x)}{z - x}.$$

Thus G_{dd} is an anti-Herglotz Cauchy transform on the upper half-plane; $M_{dd} = -G_{dd} = \langle d, (H - z)^{-1} d\rangle$ is the standard Herglotz function. In particular the density has the minus sign below.

For $|E| < 2J$, set $s(E) = \sqrt{4J^2 - E^2}$. The upper boundary value is

$$G_{dd}(E + \mathrm{i}0) = \frac{1}{E - \varepsilon + \mathrm{i}g^2/s(E)}.$$

Stone–Stieltjes inversion gives the positive density

$$\rho_d(E) = -\frac{1}{\pi} \operatorname{Im} G_{dd}(E + \mathrm{i}0) = \frac{g^2 s(E)}{\pi\{(E - \varepsilon)^2 s(E)^2 + g^4\}}. \tag{8}$$

Here is the measure-theoretic closure of this formula. Fix $K \Subset (-2J, 2J)$. The chosen square root has a continuous upper boundary value on K, and the denominator in the preceding display has nonzero imaginary part $g^2/s(E)$. Hence $G_{dd}(E + \mathrm{i}\eta) \to G_{dd}(E + \mathrm{i}0)$ uniformly on K. For every $\phi \in C_c(K)$, Stone's formula, followed by this uniform limit, gives

$$\int \phi \, d\mu_d = \lim_{\eta \downarrow 0} \int \phi(E)[-\frac{1}{\pi} \operatorname{Im} G_{dd}(E + \mathrm{i}\eta)] \, dE = \int \phi(E) \rho_d(E) \, dE. \tag{9}$$

Compact exhaustion therefore identifies the full open-band measure, not only its absolutely continuous part.

On each real exterior, G_{dd} is real analytic except at the unique simple physical pole proved in Theorem 1. Formula (9) on pole-free exterior intervals gives zero measure there, and the two poles give precisely two atoms. Finally, for $E_0 = \pm 2J$,

$$G_{dd}(z) = \frac{S(z)}{(z - \varepsilon)S(z) - g^2} \longrightarrow 0, \qquad \mu_d(\{E_0\}) = \lim_{\eta \downarrow 0} \mathrm{i}\eta G_{dd}(E_0 + \mathrm{i}\eta) = 0.$$

Any measure supported on the two leftover edge points would be atomic. Thus the density, the two exterior atoms, and the zero edge masses exhaust μ_d; no singular continuous component or embedded atom remains.

At an exterior pole E_*, differentiation of (4) gives

$$w_* = \operatorname*{Res}_{z = E_*} G_{dd}(z) = \frac{1}{1 - g^2 m'(E_*)}. \tag{10}$$

On both physical exteriors $m'(E_*) < 0$, hence $0 < w_* < 1$. Moreover

$$G_{dd}(z) = \frac{1}{z} + \frac{\varepsilon}{z^2} + \frac{\varepsilon^2 + g^2}{z^3} + \frac{\varepsilon^3 + 2\varepsilon g^2}{z^4} + O(z^{-5}). \tag{11}$$

The coefficient of z^{-1} in the Cauchy transform is the total mass, so

$$\int_{-2J}^{2J} \rho_d(E) \, dE + w_- + w_+ = 1. \tag{12}$$

Theorem 2 (Complete spectral type)**.** *For $J > 0$ and $g \neq 0$,*

$$\sigma(H) = [-2J, 2J] \cup \{E_-, E_+\}, \qquad \sigma_{\mathrm{sc}}(H) = \varnothing.$$

The band is purely absolutely continuous of multiplicity two almost everywhere; $E_\pm$ are simple atoms with weights (10) in the impurity measure. Neither band edge is an eigenvalue.

Proof. The odd component contributes one simple absolutely continuous copy of the band. Cyclicity, the explicit scalar measure above, and Theorem 1 give one further simple absolutely continuous copy and exactly the two simple atoms in the even component. At $E = \pm 2J$, the free recurrence has affine rather than decaying half-line solutions. An ℓ^2 solution must vanish on both tails and then equations (1)–(2) force all remaining coordinates to vanish. Thus there is no edge eigenvector. □

3 Stationary scattering and the Fano zero

Let $E = 2J\cos k$ with $0 < k < \pi$. Away from $E = \varepsilon$, eliminating u_d gives the on-shell origin potential

$$V(E) = \frac{g^2}{E - \varepsilon}.$$

Matching an incident wave, its reflection, and its transmission at the origin gives, in a consistent phase convention,

$$t(E) = \frac{2\mathrm{i}J\sin k}{2\mathrm{i}J\sin k + V(E)}, \qquad r(E) = \frac{-V(E)}{2\mathrm{i}J\sin k + V(E)}. \tag{13}$$

Therefore

$$T(E) = \frac{4J^2 \sin^2 k\,(E-\varepsilon)^2}{g^4 + 4J^2\sin^2 k\,(E-\varepsilon)^2}, \tag{14}$$

$$R(E) = \frac{g^4}{g^4 + 4J^2\sin^2 k\,(E-\varepsilon)^2}, \qquad T(E) + R(E) = 1. \tag{15}$$

At $E = \varepsilon$, elimination is invalid, but the original impurity equation gives $gu_0 = 0$. Thus $u_0 = 0$ and $T = 0$ directly. Consequently a continuum Fano zero occurs at $E = \varepsilon$ if and only if $|\varepsilon| < 2J$; equality is a closed channel edge, not an interior antiresonance.

The phrase *spectral scattering atlas owner* marks the first substantive revision: full spectral type, density, residues, mass, amplitudes, and antiresonance have been added to the branch-safe pole theorem.